Course 6024D | Semester VI | Theory | 4 Credits

Renewable Energy Technologies

A complete syllabus-aligned handbook with clear explanations, practical or exam guidance, Malayalam-support notes, diagrams, activities and revision tools.

Programmes	**Mechanical Engineering**
Contact hours	**60**
Teaching scheme	**3:1:0:0**
Credits	**4**
CIE / ESE	**Theory CIE 40 / Theory ESE 60**
Self-learning	**5.5 h/week**

CURRICULUM INTEGRITY

Official Source Declaration and Verified Inconsistencies

The attached Revision 2026 index identifies Course 6024D as Renewable Energy Technologies in Semester VI for Mechanical Engineering. The official SITTTR detailed course page was checked at the indexed link and returned “Requested file not found.” With the user's approved public-source approach, this handbook is transparently based on NPTEL IIT Guwahati’s Renewable Energy Engineering course and polytechnic renewable energy curricula. It does not claim to reproduce official REV2026 detailed-syllabus wording.

Source treatment: Teaching scheme, credits and assessment values are provisional placeholders aligned to neighbouring handbook format. Power plant designs, equipment specifications, energy audits and resource assessments must be based on approved engineering designs, certified manufacturer data, current safety codes and competent professional review.

Verified source note: Verified sources support solar radiation, collectors, thermal storage, biomass characterization, biogas plants, wind turbine design and power analysis. The four-module sequence is a learning path, not an asserted official module split.

COURSE DASHBOARD

What You Are Expected to Learn

60
Contact hours

4
Credits

Theory CIE 40
CIE

Theory ESE 60
ESE

Course introduction

Renewable Energy Technologies studies the engineering and design of systems that harness energy from non-conventional sources. It develops the ability to analyze solar radiation, design concentrating and non-concentrating collectors, evaluate biomass conversion routes (biochemical and thermochemical), understand wind turbine mechanics and power generation, and coordinate small hydro and geothermal systems while respecting the technical and economic boundaries of energy engineering.

> **Malayalam support:** □ handbook □□□□□□□□□□□□ syllabus topic □□□□□ □□□□□□□□□□; □□□□□□□□ principle, diagram/procedure, application, common mistake □□□□□□ □□□□□□□□□ □□□□□□□□□.

Prerequisite foundation

Thermodynamics, fluid mechanics, engineering materials, heat transfer, engineering mathematics and environmental awareness.

Use prerequisites only as a revision bridge. The current course outcomes and detailed syllabus define the required performance.

Teaching and assessment scheme

L	T	P	S	Self-learning/week	Contact hours	Credits	CIE	ESE
3	1	0	0	5.5 h/week	60	4	Theory CIE 40	Theory ESE 60

STUDY METHOD

How to Use This Handbook

1 · Understand

Read terms, conditions and purpose; make a short definition in your own words.

2 · Visualise

Follow the diagram, circuit, flow or procedure and label it accurately.

3 · Apply

Use the method block, calculation or practical checklist without skipping units or safety checks.

4 · Revise

Use questions, quick cards and common-mistake notes before examination.

OFFICIAL STATEMENTS

Objectives and Course Outcomes

Official course objectives

1. Explain solar energy fundamentals, including radiation analysis and the design of solar thermal collectors.
2. Explain biomass characterization, conversion processes (biogas, alcohol, torrefaction) and bio-digester design.
3. Explain wind energy principles, turbine types, power extraction theories and the characteristics of wind power generation.
4. Explain hydropower and geothermal energy systems, including turbine selection, pumped storage and geothermal power cycles.

Student-friendly meaning

You should be able to recognise the relevant system or material, explain its principle, communicate through a correct diagram/table where needed and follow a defensible solution or practical method.

> **Malayalam support:** CO □□□□□□□ exam-□ □□□□□ □□□□□□□□ □□□□□□□ □□□□□ □□□□□□□□□. Define, explain, apply □□□□□ action words □□□□□□□□□□□□□.

Official Course Outcomes

CO	Official outcome	Bloom level
CO1	Explain solar radiation measurement and the design of non-concentrating and concentrating collectors.	Understand
CO2	Explain biomass characterization and the biochemical/thermochemical routes for bio-energy production.	Understand
CO3	Explain wind turbine mechanics, power generation analysis and the factors affecting wind energy.	Understand
CO4	Explain the principles of hydropower, geothermal energy and energy storage technologies.	Understand

Official CO-PO articulation matrix

CO	PO1	PO2	PO3	PO4	PO5	PO6	PO7–PO11
CO1	3	2	2	2	2	–	–
CO2	3	2	2	2	2	–	–
CO3	3	2	2	2	2	–	–
CO4	3	2	2	2	2	–	–

SYLLABUS COVERAGE

Curriculum Coverage Map

All official major topics mapped to handbook chapters					
Module	**Title**	**Official major topics**	**Hours**	**CO**	**Handbook section**
1	Solar Thermal Engineering and Storage	Solar radiation basics; measurement of solar energy; non-concentrating collectors (flat plate); concentrating collectors (parabolic, heliostat); thermal energy storage systems; solar energy utilization methods.	Approx. 14 hours	CO1	Module 1
2	Biomass Characterization and Conversion	Classification of biomass resources; compositional analysis (proximate and ultimate); biochemical conversion (biogas, fermentation); thermochemical conversion (torrefaction, pyrolysis, gasification, combustion).	Approx. 16 hours	CO2	Module 2
3	Wind Energy Mechanics and Generation	Wind energy basics; wind turbine terms and types (HAWT, VAWT); Betz limit and power extraction theory; turbine design and performance; wind speed analysis and power generation characteristics.	Approx. 16 hours	CO3	Module 3
4	Hydro, Geothermal and Integrated Systems	Hydropower principles; hydroturbine selection; small and micro hydro plants; pumped hydro storage; geothermal energy sources and power cycles; integrated renewable energy systems and energy storage.	Approx. 14 hours	CO4	Module 4

LEARNING ROADMAP

How the Modules Connect

Solar Thermal Engineering and Storage

Solar radiation basics; measurement of solar energy; non-concentrating collectors (flat plate); concentrating collectors (parabolic, heliostat); thermal energy storage systems; solar energy utilization methods.

CO1 · Approx. 14 hours

Biomass Characterization and Conversion

Classification of biomass resources; compositional analysis (proximate and ultimate); biochemical conversion (biogas, fermentation); thermochemical conversion (torrefaction, pyrolysis, gasification, combustion).

CO2 · Approx. 16 hours

Wind Energy Mechanics and Generation

Wind energy basics; wind turbine terms and types (HAWT, VAWT); Betz limit and power extraction theory; turbine design and performance; wind speed analysis and power generation characteristics.

CO3 · Approx. 16 hours

Hydro, Geothermal and Integrated Systems

Hydropower principles; hydroturbine selection; small and micro hydro plants; pumped hydro storage; geothermal energy sources and power cycles; integrated renewable energy systems and energy storage.

CO4 · Approx. 14 hours

MODULE 1

Solar Thermal Engineering and Storage

Solar radiation basics; measurement of solar energy; non-concentrating collectors (flat plate); concentrating collectors (parabolic, heliostat); thermal energy storage systems; solar energy utilization methods.

Approx. 14 hours **CO1** **Understand · Apply**

Learning goals

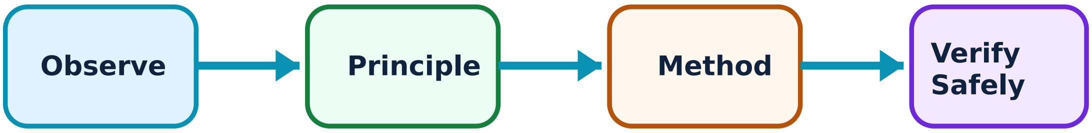
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Solar Thermal Engineering and Storage: identify the system, state its principle, follow the correct method and verify the result safely.

1. Solar radiation and flat plate collectors−

Solar radiation is the primary energy source. Measurement involves global, diffuse and beam radiation. Flat plate collectors are non-concentrating devices used for low-temperature applications like water heating, where heat is absorbed by a blackened plate and transferred to a fluid.

Study and application method

List the components of a flat plate collector and explain how it differs from a concentrating collector; define 'global solar radiation'.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List the components of a flat plate collector and explain how it differs from a concentrating collector; define 'global solar radiation'.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□.

Common mistake / precaution: Solar collector efficiency is sensitive to orientation and surface coating. Do not design systems without authorised solar-path analysis and local weather data.

2. Concentrating collectors and storage−

Concentrating collectors (like parabolic troughs) focus solar energy onto a small receiver to achieve high temperatures for power generation. Thermal energy storage (sensible and latent heat) allows solar energy to be used when the sun is not shining, improving system reliability.

Study and application method

Sketch a parabolic trough collector and label the receiver; compare sensible heat storage and latent heat storage for solar thermal systems.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Sketch a parabolic trough collector and label the receiver; compare sensible heat storage and latent heat storage for solar thermal systems.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Concentrating systems involve high temperatures and tracking mechanisms. Do not operate these systems without authorised safety training and automatic control systems.

Module 1 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 2

Biomass Characterization and Conversion

Classification of biomass resources; compositional analysis (proximate and ultimate); biochemical conversion (biogas, fermentation); thermochemical conversion (torrefaction, pyrolysis, gasification, combustion).

Approx. 16 hours | **CO2** | **Understand · Apply**

Learning goals

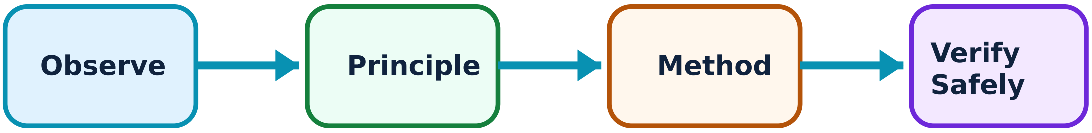
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Biomass Characterization and Conversion: identify the system, state its principle, follow the correct method and verify the result safely.

1. Biomass properties and biochemical conversion−

Biomass includes agricultural residues, wood and organic waste. Biochemical conversion uses microorganisms to produce biogas (anaerobic digestion) or alcohol (fermentation). Biogas digesters are designed based on the substrate type and required gas output.

Study and application method

Describe the stages of anaerobic digestion in a biogas plant; list four characteristics used to classify biomass resources.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Describe the stages of anaerobic digestion in a biogas plant; list four characteristics used to classify biomass resources.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Biogas plants involve biological and flammable gas hazards. Do not operate digesters without authorised ventilation and pressure-relief systems.

2. Thermochemical conversion processes−

Thermochemical routes use heat to convert biomass into solid (char), liquid (bio-oil) or gaseous (syngas) fuels. Processes include torrefaction (low heat), pyrolysis (no oxygen), gasification (partial oxygen) and direct combustion.

Study and application method

Compare the products of biomass pyrolysis and gasification; explain the purpose of 'torrefaction' in biomass processing.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Compare the products of biomass pyrolysis and gasification; explain the purpose of 'torrefaction' in biomass processing.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Thermochemical processes involve high temperatures and toxic/flammable gases. All equipment must meet authorised industrial safety standards for pressure and temperature.

Module 2 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 3

Wind Energy Mechanics and Generation

Wind energy basics; wind turbine terms and types (HAWT, VAWT); Betz limit and power extraction theory; turbine design and performance; wind speed analysis and power generation characteristics.

Approx. 16 hours | **CO3** | **Understand · Apply**

Learning goals

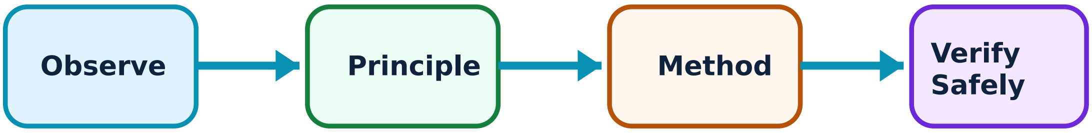
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Wind Energy Mechanics and Generation: identify the system, state its principle, follow the correct method and verify the result safely.

1. Wind turbine principles and types−

Wind turbines convert kinetic energy of wind into mechanical power. Horizontal Axis Wind Turbines (HAWT) are common for large-scale generation. The Betz limit defines the maximum theoretical efficiency (59.3%) of any wind turbine.

Study and application method

Explain the 'Betz limit' in wind energy extraction; compare the advantages of HAWT and VAWT (Vertical Axis Wind Turbine).

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Explain the 'Betz limit' in wind energy extraction; compare the advantages of HAWT and VAWT (Vertical Axis Wind Turbine).

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Wind turbines involve high mechanical loads and rotating masses. Do not plan turbines in high-wind areas without authorised structural and fatigue analysis.

2. Wind power analysis and performance−

Power generation depends on wind speed (cubic relationship) and turbine diameter. Performance is analyzed using power curves, cut-in speeds and rated speeds. Wind speed analysis (Weibull distribution) is critical for predicting annual energy production.

Study and application method

Calculate the potential power in the wind conceptually for a given speed and diameter; describe the 'cut-in' and 'cut-out' speeds of a wind turbine.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Calculate the potential power in the wind conceptually for a given speed and diameter; describe the 'cut-in' and 'cut-out' speeds of a wind turbine.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Wind speed data is highly site-specific. Do not rely on generic wind maps for project-specific investment decisions without authorised site measurements.

Module 3 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

MODULE 4

Hydro, Geothermal and Integrated Systems

Hydropower principles; hydroturbine selection; small and micro hydro plants; pumped hydro storage; geothermal energy sources and power cycles; integrated renewable energy systems and energy storage.

Approx. 14 hours | **CO4** | **Understand · Apply**

Learning goals

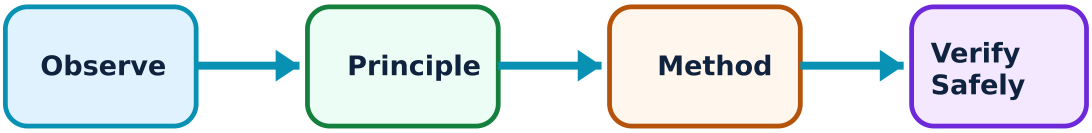
Identify core terms, explain the principle and complete the stated application or practical outcome.

Key evidence

Use a labelled diagram, table, graph, procedure or calculation that matches the topic.

Engineering habit

Check units, polarity/direction, ratings, materials and safety before concluding.



Study flow for Hydro, Geothermal and Integrated Systems: identify the system, state its principle, follow the correct method and verify the result safely.

1. Hydropower and geothermal systems−

Hydropower uses water flow to drive turbines, with selection (Pelton, Francis, Kaplan) based on head and flow. Geothermal energy harnesses Earth's heat through steam or binary cycles. Both provide stable, base-load renewable power.

Study and application method

Identify the criteria for selecting a Pelton wheel turbine; describe the working of a 'binary cycle' geothermal power plant.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: Identify the criteria for selecting a Pelton wheel turbine; describe the working of a 'binary cycle' geothermal power plant.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

Malayalam support: □□□□□ concept □□□□□□□ □□□□□ □□□□□□□□□□□□□□. □□□□ diagram / circuit / formula / procedure □□□□□□□□□□□ □□□□□□□□. □□□□□□□□□ result □□□□□□□□□□ application □□□□□□□ □□□□□□□□□□ □□□□□ □□□□□□□□□□□□□. Technical terms English-□ □□□□□ □□□□□□□□□□□□.

Common mistake / precaution: Hydro and geothermal projects have significant environmental and geological impacts. All projects must undergo authorised environmental and seismic impact assessments.

2. Energy storage and hybrid systems−

Renewables are often intermittent. Energy storage (batteries, pumped hydro, flywheels) and hybrid systems (e.g., solar-wind) ensure a continuous power supply. Integrated systems optimize resource use and improve grid stability.

Study and application method

List three energy storage technologies suitable for renewable energy systems; explain the benefits of a hybrid solar-wind power system.

Connect the explanation to the official student learning outcome before attempting a numerical, diagram, procedure or viva answer.

Evidence of understanding

Use correct terminology, a labelled sketch/table where useful, correct units or conditions, and a conclusion linked to the observed or calculated result.

How to construct a complete answer

Given: identify the quantities, device condition, waveform or circuit arrangement.

Required: state exactly what must be found or explained.

Method: List three energy storage technologies suitable for renewable energy systems; explain the benefits of a hybrid solar-wind power system.

Answer habit: show a labelled step, the relevant unit/condition and a short interpretation.

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Common mistake / precaution: Energy storage systems involve chemical, mechanical or thermal hazards. All storage facilities must meet authorised safety and grid-integration standards.

Module 4 revision cue: Re-read official major topics, redraw one key diagram/flow and explain one practical or numerical method without looking at notes.

FORMULA AND PROCEDURE BANK

Rapid Recall Bank

Recall 1

State symbols and SI units before substitution.

Recall 2

Draw the circuit, system boundary, vector or process flow before reasoning.

Recall 3

For laboratory work: aim → apparatus → diagram → procedure → observation → calculation → result → precautions.

Recall 4

For a graph: label axes, units, scale, operating region and conclusion.

Recall 5

For a comparison: use construction, working, result, advantage and limitation.

Recall 6

For an extended answer: definition/principle, labelled diagram, working steps, application and a caution/limitation.

DIAGRAM LIBRARY

Diagram Library for Rapid Revision

Module 1: Solar Thermal Engineering and Storage

Use the concept flow to revise observation, principle, method and verification.

Module 2: Biomass Characterization and Conversion

Use the concept flow to revise observation, principle, method and verification.

Module 3: Wind Energy Mechanics and Generation

Use the concept flow to revise observation, principle, method and verification.

Module 4: Hydro, Geothermal and Integrated Systems

Use the concept flow to revise observation, principle, method and verification.

SELF-LEARNING

Official Self-Learning and Student Activities

Structured activity

Compare solar, wind and biomass energy in a conceptual table showing their energy density and typical applications.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Sketch a cross-section of a flat plate collector and label the heat transfer path.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Draw a flow chart for the thermochemical conversion of biomass to syngas.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Calculate the theoretical maximum efficiency of a wind turbine conceptually using the Betz limit.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Structured activity

Identify the key components of a small-scale hydropower plant and label them.

Keep evidence in your record: date, source/observation, diagram/table, key learning and questions for discussion.

Activity discipline: The supplied PDF assigns the listed self-learning hours. This handbook preserves the activity direction; the teacher’s approved schedule and safety instructions govern execution.

EXAMINATION READINESS

Assessment Methodology and Examination Pattern

Official assessment structure

Component	Official mark / conversion	Student evidence
Attendance	5 marks	End-of-semester attendance component.
Self-learning activities	15 marks	Module-linked activities.
Series examinations	20 + 20 converted	Two 50-mark series examinations.
CIE total	Theory CIE 40	Continuous internal evaluation.
Written ESE	Theory ESE 60	2.5 hours; official Part A / Part B / Part C pattern.

Series-test pattern

The supplied theory pattern uses Part A (1-mark), Part B (3-mark with choice) and Part C (7-mark) distribution across the stated modules. Practical courses follow the supplied practical-test scheme.

Examination evidence

Show a complete method, correct units/conditions, clear diagrams or tables, a result/conclusion and safety awareness for any apparatus or process involved.

EXAM PRACTICE

Module-wise Questions with Answers

Module 1 — Solar Thermal Engineering and Storage

Explain Solar radiation and flat plate collectors and state one correct method, application or precaution.−

Solar radiation is the primary energy source. Measurement involves global, diffuse and beam radiation. Flat plate collectors are non-concentrating devices used for low-temperature applications like water heating, where heat is absorbed by a blackened plate and transferred to a fluid.

Answer framework: List the components of a flat plate collector and explain how it differs from a concentrating collector; define 'global solar radiation'.

Important point: Solar collector efficiency is sensitive to orientation and surface coating. Do not design systems without authorised solar-path analysis and local weather data.

Explain Concentrating collectors and storage and state one correct method, application or precaution.−

Concentrating collectors (like parabolic troughs) focus solar energy onto a small receiver to achieve high temperatures for power generation. Thermal energy storage (sensible and latent heat) allows solar energy to be used when the sun is not shining, improving system reliability.

Answer framework: Sketch a parabolic trough collector and label the receiver; compare sensible heat storage and latent heat storage for solar thermal systems.

Important point: Concentrating systems involve high temperatures and tracking mechanisms. Do not operate these systems without authorised safety training and automatic control systems.

Module 2 — Biomass Characterization and Conversion

Explain Biomass properties and biochemical conversion and state one correct method, application or precaution.−

Biomass includes agricultural residues, wood and organic waste. Biochemical conversion uses microorganisms to produce biogas (anaerobic digestion) or alcohol (fermentation). Biogas digesters are designed based on the substrate type and required gas output.

Answer framework: Describe the stages of anaerobic digestion in a biogas plant; list four characteristics used to classify biomass resources.

Important point: Biogas plants involve biological and flammable gas hazards. Do not operate digesters without authorised ventilation and pressure-relief systems.

Explain Thermochemical conversion processes and state one correct method, application or precaution.−

Thermochemical routes use heat to convert biomass into solid (char), liquid (bio-oil) or gaseous (syngas) fuels. Processes include torrefaction (low heat), pyrolysis (no oxygen), gasification (partial oxygen) and direct combustion.

Answer framework: Compare the products of biomass pyrolysis and gasification; explain the purpose of 'torrefaction' in biomass processing.

Important point: Thermochemical processes involve high temperatures and toxic/flammable gases. All equipment must meet authorised industrial safety standards for pressure and temperature.

Module 3 — Wind Energy Mechanics and Generation

Explain Wind turbine principles and types and state one correct method, application or precaution.−

Wind turbines convert kinetic energy of wind into mechanical power. Horizontal Axis Wind Turbines (HAWT) are common for large-scale generation. The Betz limit defines the maximum theoretical efficiency (59.3%) of any wind turbine.

Answer framework: Explain the 'Betz limit' in wind energy extraction; compare the advantages of HAWT and VAWT (Vertical Axis Wind Turbine).

Important point: Wind turbines involve high mechanical loads and rotating masses. Do not plan turbines in high-wind areas without authorised structural and fatigue analysis.

Explain Wind power analysis and performance and state one correct method, application or precaution.−

Power generation depends on wind speed (cubic relationship) and turbine diameter. Performance is analyzed using power curves, cut-in speeds and rated speeds. Wind speed analysis (Weibull distribution) is critical for predicting annual energy production.

Answer framework: Calculate the potential power in the wind conceptually for a given speed and diameter; describe the 'cut-in' and 'cut-out' speeds of a wind turbine.

Important point: Wind speed data is highly site-specific. Do not rely on generic wind maps for project-specific investment decisions without authorised site measurements.

Module 4 — Hydro, Geothermal and Integrated Systems

Explain Hydropower and geothermal systems and state one correct method, application or precaution.−

Hydropower uses water flow to drive turbines, with selection (Pelton, Francis, Kaplan) based on head and flow. Geothermal energy harnesses Earth's heat through steam or binary cycles. Both provide stable, base-load renewable power.

Answer framework: Identify the criteria for selecting a Pelton wheel turbine; describe the working of a 'binary cycle' geothermal power plant.

Important point: Hydro and geothermal projects have significant environmental and geological impacts. All projects must undergo authorised environmental and seismic impact assessments.

Explain Energy storage and hybrid systems and state one correct method, application

or precaution.–

Renewables are often intermittent. Energy storage (batteries, pumped hydro, flywheels) and hybrid systems (e.g., solar-wind) ensure a continuous power supply. Integrated systems optimize resource use and improve grid stability.

Answer framework: List three energy storage technologies suitable for renewable energy systems; explain the benefits of a hybrid solar-wind power system.

Important point: Energy storage systems involve chemical, mechanical or thermal hazards. All storage facilities must meet authorised safety and grid-integration standards.

PRACTICE ONLY

Practice Model Paper

Important: This practice paper is based on the official pattern in the supplied syllabus. It is **not** an official SITTTR model question paper.

Part A — short response

1. State one key definition from Module 1: Solar Thermal Engineering and Storage.
2. Identify one diagram, observation, formula or safety condition relevant to Module 1.
3. State one key definition from Module 2: Biomass Characterization and Conversion.
4. Identify one diagram, observation, formula or safety condition relevant to Module 2.
5. State one key definition from Module 3: Wind Energy Mechanics and Generation.
6. Identify one diagram, observation, formula or safety condition relevant to Module 3.
7. State one key definition from Module 4: Hydro, Geothermal and Integrated Systems.
8. Identify one diagram, observation, formula or safety condition relevant to Module 4.

Part B — structured explanation

1. Explain a selected procedure/principle from Module 1 with a labelled diagram, table or method.
2. Explain a selected procedure/principle from Module 2 with a labelled diagram, table or method.
3. Explain a selected procedure/principle from Module 3 with a labelled diagram, table or method.
4. Explain a selected procedure/principle from Module 4 with a labelled diagram, table or method.

Part C — extended response / practical task

1. Develop a complete answer or practical plan for: Solar radiation basics; measurement of solar energy; non-concentrating collectors (flat plate); concentrating collectors (parabolic, heliostat); thermal energy storage systems; solar energy utilization methods..

2. Develop a complete answer or practical plan for: Classification of biomass resources; compositional analysis (proximate and ultimate); biochemical conversion (biogas, fermentation); thermochemical conversion (torrefaction, pyrolysis, gasification, combustion)..
3. Develop a complete answer or practical plan for: Wind energy basics; wind turbine terms and types (HAWT, VAWT); Betz limit and power extraction theory; turbine design and performance; wind speed analysis and power generation characteristics..
4. Develop a complete answer or practical plan for: Hydropower principles; hydroturbine selection; small and micro hydro plants; pumped hydro storage; geothermal energy sources and power cycles; integrated renewable energy systems and energy storage..

LAST-MILE REVISION

Quick Revision

Module 1: Solar Thermal Engineering and Storage

Solar radiation basics; measurement of solar energy; non-concentrating collectors (flat plate); concentrating collectors (parabolic, heliostat); thermal energy storage systems; solar energy utilization methods.

- Match each topic to CO1.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 2: Biomass Characterization and Conversion

Classification of biomass resources; compositional analysis (proximate and ultimate); biochemical conversion (biogas, fermentation); thermochemical conversion (torrefaction, pyrolysis, gasification, combustion).

- Match each topic to CO2.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Module 3: Wind Energy Mechanics and Generation

Wind energy basics; wind turbine terms and types (HAWT, VAWT); Betz limit and power extraction theory; turbine design and performance; wind speed analysis and power generation characteristics.

- Match each topic to CO3.
- Redraw or rehearse one key method.

- Check one common mistake/precaution.

Module 4: Hydro, Geothermal and Integrated Systems

Hydropower principles; hydroturbine selection; small and micro hydro plants; pumped hydro storage; geothermal energy sources and power cycles; integrated renewable energy systems and energy storage.

- Match each topic to CO4.
- Redraw or rehearse one key method.
- Check one common mistake/precaution.

Common mistakes: Writing an unlabelled diagram; ignoring units, polarity or conditions; treating a practical result as valid without observations; confusing a definition with an application; and omitting a safety precaution where the topic uses apparatus, current, heat, chemicals or tools.

Exam checklist: Read the command word; answer every part; draw clean labelled diagrams; use a clear calculation/procedure; state result with units or observation; and reserve two minutes for checking.

VOCABULARY

Glossary

Important terms from the syllabus	
Term / topic	**One-line meaning**
Solar radiation and flat plate collectors	Solar radiation is the primary energy source.
Concentrating collectors and storage	Concentrating collectors (like parabolic troughs) focus solar energy onto a small receiver to achieve high temperatures for power generation.
Biomass properties and biochemical conversion	Biomass includes agricultural residues, wood and organic waste.
Thermochemical conversion processes	Thermochemical routes use heat to convert biomass into solid (char), liquid (bio-oil) or gaseous (syngas) fuels.
Wind turbine principles and types	Wind turbines convert kinetic energy of wind into mechanical power.
Wind power analysis and performance	Power generation depends on wind speed (cubic relationship) and turbine diameter.
Hydropower and geothermal systems	Hydropower uses water flow to drive turbines, with selection (Pelton, Francis, Kaplan) based on head and flow.
Energy storage and hybrid systems	Renewables are often intermittent.

LEARNING RESOURCES

References

Recommended renewable-technology references

1. V. V. Goud and R. Anandalakshmi, Renewable Energy Engineering.
2. S. P. Sukhatme, Solar Energy.
3. G. D. Rai, Non-Conventional Energy Sources.
4. J. Twidell and T. Weir, Renewable Energy Resources.

Approved public renewable-technology sources

- https://nptel.ac.in/courses/103103206
- https://nptel.ac.in/courses/112106318

These sources provide the verified study basis while official Course 6024D detail is unavailable; they do not replace official engineering designs, certified equipment specifications, official energy audits or authorised production plans.